Atlas, carries and rank: implementation follow-up

2026-11-04. All check outcomes below are producer observations with **unverified standing** [pcc:unverified]. The original study, claim, integer failures and published branch remain unchanged. This follow-up completes the native runner, bounded solver representation, arithmetic replay and source-bound packaging.

Requested work and evidence

Requirement	Implemented deliverable	Remaining acceptance work
1. Isogeny representatives	Five matched support arms; RFC-parameter GMP fixture; small-prime homomorphism, dual and inverse checks	Original admitted panel and matched solver costs
2. Sparse addition and carries	Full signed, distinct-column three-sum constraints; bounded bit-vector lowering of mod and carry encodings	Complete fixed-cell comparisons, including censored outcomes
3. Useful relation selection	Training-only coefficient-rank/probe criterion; holdout sign orbits disjoint from training; raw generic rows retained	Held-out usefulness and complete selection/setup accounting
Native execution	Write-once receipts, input/binary hashes, explicit failures, independent OpenSSL arithmetic implementation	EKS execution and independent signed verification
Reproducible packaging	Published source, locked Rust dependencies, digest-pinned base images, UID 1000 runtime	Registry digest, checkpoint and admitted deployment receipts

These are compositions of existing leads, not established novel mechanisms. The original source review records RFC 7748, Galbraith chapter 15, the lab lead atlas, existing Edwards systems, traits and the tower-oracle gap. Before this extension, the SMT-LIB fixed bit-vector theory and native OpenSSL EC/EVP interfaces were also checked. The bit-vector work changes the representation of bounded equations; it does not supply compact factor-base membership or Weil descent.

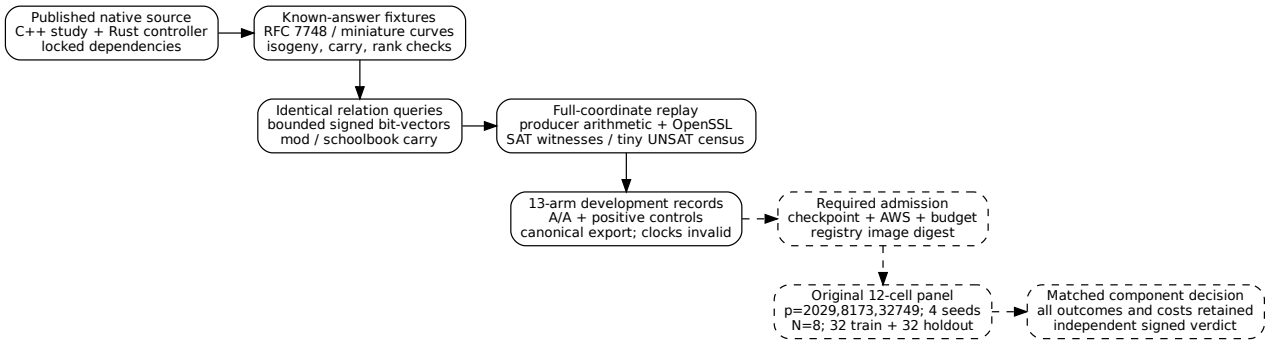

Figure 1: Implementation and outstanding admission

Editable diagram: DOT; vector: SVG. Dashed stages are outstanding. This is a workflow diagram, not an isogeny graph or performance plot.

Bounded solver representation

`bv_lowering.hpp` lowers the bounded integer expression grammar to signed bit-vectors. The width is `max(20, 2*field_bits+5)` in the supported miniature domain. Products, linear combinations and radix carry columns have headroom; canonical field variables remain explicitly bounded. Signed remainder is corrected by adding p when negative. The retained fixture checks exercise the actual solver formulas, including deliberately incorrect residues.

The fixture records retain 196 product cases and 30 relation cases: 20 SAT models replayed natively, 10 UNSAT answers matching the finite reference [pcc:unverified]. Each archive is below 64 KiB; manifest and summary files remain beside them. Earlier QF_NIA timeouts stay in the original checks. No speed ratio is inferred between these separate development attempts.

Runner, replay and records

`control/` is a native Rust controller. Each fresh solver process retains its arguments, raw stdout/stderr, resource counter, exit status, witness replay and per-row receipt. It accepts the precise Z3 get-model-after-UNSAT diagnostic; other solver errors invalidate the row. Timeouts are neither UNSAT nor wins.

`openssl_replay.cpp` uses OpenSSL point arithmetic rather than the producer's group-law implementation. It converts Montgomery coordinates to a short Weierstrass model, checks subgroup/base membership and verifies the complete signed three-point sum. Tiny UNSAT cases use a separate exhaustive OpenSSL census. The RFC fixture also cross-checks three clamped X25519 outputs using OpenSSL EVP. These producer-invoked checks do not establish independent execution or PCC verification [pcc:unverified].

The retained OpenSSL rejection checks reject a forged UNSAT result for a known positive query and a duplicate model coordinate [pcc:unverified]. Seven export canaries reject changed raw output, mismatched receipts, forged resource counts, wrong source, wrong codec, changed column count and a changed deadline [pcc:unverified].

Development record	Planned / attempted	Successful producer checks	Retained limitation
Historical `unit-example`	13 / 13	13	Predates mandatory external replay/source binding
`unit-example2`	13 / 13	10	Three carry queries timed out at 30 seconds
`image-smoke`	13 / 13	13	Local container integration only
`final-image-smoke`	13 / 13	13	Final native/exporter packaging; local integration only

All table observations are [pcc:unverified]. `metadata-recheck` repeats the canonical export/file check with the tightened exporter and the same retained local records; it is not another solver run. Canonical schema acceptance does not promote the recorded scientific or execution standing. Every export uses `verification.status=no_claim`; clocks remain null and invalid. Z3 resource units exclude generation, transport, selection, replay and matrix work and are not GAE, S, wall time or complete pipeline cost.

Exact source and packaging

The first full image smoke uses published source `ecb2e20aac22d99fd628f4c8b4861dcd57645135`. `image-smoke` records the image configuration, four native binary hashes, clean-source input configuration, 13 row receipts and canonical export. `validation/image.iid` is the local Docker image identifier, **not** an ECR manifest digest.

The final published build source is `5a7e0fa433aa01dfefee8ec7fcb7ad2e6765a957`: it includes tightened export identity, deadline, codec and column bindings and fewer builder layers for VFS storage. Its container receipt is retained separately in `final-image-smoke`; source and image identities must not be combined across the two runs. An intermediate rebuild failed for lack of disk space before any fixture ran; its log and cache cleanup receipt remain in `validation`.

The image is built from an exact committed study subtree. It contains native Z3 5.3.3, the C++ study and OpenSSL replay executable, and the locked Rust controller. A build-only CA secret supplies the execution environment's network trust; it is not embedded in the image. The runtime runs as UID 1000, and local smoke execution disables networking.

Original panel and real blockers

Twelve prospective protocol drafts preserve fields 2029/8173/32749, seeds 13/29/47/71, N=8 and 32 training plus 32 held-out targets. Each cell has 416 solver rows; the panel plans 4,992 rows. Schema checks pass, but draft acceptance is not admission [pcc:unverified]. Runtime preflight is unmeasured, source/image pins are not frozen for deployment, and the USD 5 aggregate budget headroom has not been established.

AWS access has no configured credentials. The checkpoint gate explicitly refuses because neither an AWS profile nor a checkpoint upload token exists. The GitHub ecbench build route builds its own instrument rather than this custom study; dispatching its stock fixture would not execute this request. No skip, IAM change, quota increase, paid job or substitute scientific run was used.

Next executable step: establish the existing authorized EKS/checkpoint access, check live capacity and budget, publish this custom image by digest, freeze the exact source/protocol/input identities, run the unchanged panel and replay every retained outcome. Any preflight-driven protocol amendment must precede affected data and remain visible. A complete comparison and signed independent verdict remain outstanding; no index-calculus improvement is claimed [pcc:unverified]. The problem record and blocked backlog card retain these exact requirements, including the missing queue/checkpoint and independent-checker integration before a ready-card promotion.

Repository validation

Native self-tests, three Rust controller tests, strict clippy and the metadata checks are in validation. The PCC compatibility fix is in crypto #3434, with matching autoresearcher #3023 and cryptanalysis #1401. Twelve PCC tests and 59 sabotage classes pass locally [pcc:unverified]. The immutable original claim remains `not_covered`; no producer signature or new trust was added.

The broad release-library check finished: **6,006 passed, 12 failed, 130 ignored** out of 6,148 declared tests, exit 101 [pcc:unverified]. The retained receipt reconciles that full attempt. Sparse device and bounds fixtures were restored, after which six device tests and the bounds test pass on targeted rerun; this repairs six of the original failure cases [pcc:unverified]. Diagnostic clock tests require `/proc/self/ns/time` and `/proc/self/timens_offsets`, absent on this host. Four untouched assertions also fail: Gaudry four-large-prime coverage, residual operation accounting, wide/word pipeline count equality, and ML profile-limit binding. The library sources and root Cargo files are byte-identical to the compiled base across this follow-up. The full suite is not a passing gate, and targeted successes do not replace it. No assertion is relaxed or ignored here. Raw log parts decompress in filename order to the original SHA-256-pinned log.

Reproduction

From the study directory, install the documented native dependencies, then:

```
make test
cargo test --locked --manifest-path control/Cargo.toml
cargo clippy --locked --manifest-path control/Cargo.toml \
  -- -D warnings
cargo build --locked --release --manifest-path control/Cargo.toml
study=/tmp/curve25519-atlas-carry-rank
solver=/path/to/z3
bash test-products.sh "$study" "$solver" /tmp/fresh-products bv
bash test-encodings.sh "$study" "$solver" /tmp/fresh-relations full bv
```

The container reproduction script binds a published commit, fresh absolute output directory and canonical checker path. It retains incomplete outcomes and exports all completed row receipts. The `--runner-fixture` command generates the fixed local integration fixture; it is not a replacement for `--run P 8 32 SEED FRESH_OUTPUT bv`. Runner/export commands, source-bound container construction and complete retained records provide development reproducibility; independent verification remains open.